

1.4 Indices

Indices are also known as powers or exponents. An index tells us how many times a quantity is multiplied by itself. For example,

$$3^4 = 3 \times 3 \times 3 \times 3 = 81$$

In the expression

$$3^4$$

the number 3 is called the base and the number 4 is called the index. Indices provide a compact way to write repeated multiplication. For example,

$$2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$$

can be written more simply as

$$2^8$$

Definition

If no index is shown, the index is 1. For example

$$5 = 5^1$$

and

$$x = x^1$$

1.4.1 Multiplying Indices

Consider

$$4^3 \times 4^5$$

Writing each power in full gives

$$(4 \times 4 \times 4) \times (4 \times 4 \times 4 \times 4 \times 4)$$

Counting the factors of 4 shows that there are eight in total:

$$4^3 \times 4^5 = 4^8$$

Notice that

$$8 = 3 + 5$$

This leads to the following rule.

Definition

When multiplying powers with the same base, add the indices.

$$a^m \times a^n = a^{m+n}$$

Worked Example

Simplify

$$2^4 \times 2^7$$

Add the indices:

$$2^{4+7} = 2^{11}$$

1.4.2 Dividing Indices

Consider

$$\frac{6^5}{6^2}$$

Writing each power in full gives

$$\frac{6 \times 6 \times 6 \times 6 \times 6}{6 \times 6}$$

Two factors of 6 cancel, leaving

$$6 \times 6 \times 6 = 6^3$$

Notice that

$$3 = 5 - 2$$

This leads to the following rule.

Definition

When dividing powers with the same base, subtract the indices.

$$\frac{a^m}{a^n} = a^{m-n}$$

Worked Example

Simplify

$$\frac{7^8}{7^3}$$

Subtract the indices:

$$7^{8-3} = 7^5$$

1.4.3 Powers of Powers

Consider the expression

$$(a^3)^2$$

Writing the inner power in full gives

$$(a \times a \times a)^2$$

This means

$$(a \times a \times a)(a \times a \times a)$$

Collecting all the factors gives

$$a^6$$

Since

$$6 = 3 \times 2$$

we obtain the following rule.

Definition

When a power is raised to another power, multiply the indices.

$$(a^m)^n = a^{mn}$$

Worked Example

Simplify

$$(5^2)^3$$

Multiply the indices:

$$5^{2 \times 3} = 5^6$$

1.4.4 Zero Indices

Consider

$$\frac{a^2}{a^2}$$

Using the division rule,

$$\frac{a^2}{a^2} = a^{2-2} = a^0$$

However,

$$\frac{a^2}{a^2} = 1$$

Therefore,

$$a^0 = 1$$

This gives the following rule.

Definition

Any non-zero number raised to the power zero is equal to one.

$$a^0 = 1$$

Worked Example

Evaluate 17^0 .

$$17^0 = 1$$

1.4.5 Negative Indices

Many students think a negative index produces a negative answer. This is not true. A negative index indicates a reciprocal. Consider

$$\frac{a^2}{a^6}$$

Using the division rule,

$$\frac{a^2}{a^6} = a^{2-6} = a^{-4}$$

However,

$$\frac{a^2}{a^6} = \frac{aa}{aaaaaa} = \frac{1}{a^4}$$

Therefore,

$$a^{-4} = \frac{1}{a^4}$$

This leads to the following rule.

Definition

A negative index indicates a reciprocal.

$$a^{-m} = \frac{1}{a^m}$$

Worked Example

Evaluate 2^{-3} .

Using the rule,

$$2^{-3} = \frac{1}{2^3} = \frac{1}{8}$$

1.4.6 Fractional Indices

Consider

$$a^{\frac{1}{2}} \times a^{\frac{1}{2}}$$

Using the multiplication rule,

$$a^{\frac{1}{2} + \frac{1}{2}} = a^1 = a$$

We also know that

$$\sqrt{a} \times \sqrt{a} = a$$

Therefore,

$$a^{\frac{1}{2}} = \sqrt{a}$$

This leads to the following rule.

Definition

$$a^{\frac{1}{n}} = \sqrt[n]{a}$$

Worked Example

Evaluate $64^{\frac{1}{3}}$.

$$64^{\frac{1}{3}} = \sqrt[3]{64} = 4$$

More generally,

$$(a^{1/n})^m = a^{m/n}$$

Hence,

Definition

$$a^{\frac{m}{n}} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m$$

Worked Example

Evaluate $25^{\frac{3}{2}}$.

$$25^{\frac{3}{2}} = (25^{\frac{1}{2}})^3 = (\sqrt{25})^3 = 5^3 = 125$$

1.4.7 Common Mistakes

Common errors involving indices include:

- Thinking $a^m + a^n = a^{m+n}$.
- Confusing $(a^m)^n$ with $a^m \times a^n$.
- Assuming a negative index gives a negative answer.
- Forgetting that $a^0 = 1$.
- Forgetting that fractional indices represent roots.
- Ignoring brackets, for example confusing $(-2)^4$ with -2^4 .

1.4.8 Engineering Application

Engineering Application

Indices occur frequently throughout engineering and science.

The area of a square plate with side length 3 m is

$$3^2 = 9 \text{ m}^2$$

The volume of a cube with side length 2 m is

$$2^3 = 8 \text{ m}^3$$

Indices also appear in scientific notation. For example,

$$3 \times 10^6$$

represents three million.

Understanding indices is therefore essential when working with formulae, measurements and engineering calculations.

Summary

Definition

For any suitable values of a , m and n ,

$$a^m \times a^n = a^{m+n}$$

$$\frac{a^m}{a^n} = a^{m-n}$$

$$(a^m)^n = a^{mn}$$

$$a^{-n} = \frac{1}{a^n}$$

$$a^0 = 1$$

$$a^{1/n} = \sqrt[n]{a}$$

$$a^{m/n} = \sqrt[n]{a^m}$$

Indices appear throughout mathematics, engineering and science. These rules should become second nature.

1.4.9 Exercises

Exercise

1. Evaluate:

(a) 7^3

(b) 3^7

(c) 5^{-2}

(d) $2^{5/2}$

(e) $(-2)^6$

(f) -2^6

2. Express each as a single power:

(a) $2^3 \times 2^5 \times 2^7$

(b) $3^3 \times 3^{-4}$

(c) $4^6 \times \frac{4^{-7}}{4^{-5}}$

(d) $\frac{5^5 \times 5^{-3}}{5^4 \times 5^{-2} \times 5^{-7}}$

(e) $(2^{-3})^4$

(f) $3^{2/3} \times 3^{-2}$

3. Simplify:

(a) $y^5 \times y^8$

(b) $\frac{x^{10}}{x^7}$

(c) $x^3 \div x$

(d) $(a^2)^3$

(e) $3x^2 \times 4x^7 \times 2x^{-3}$

(f) $6x \times 7y^2 \times x^5$

4. Change to the specified base:

(a) 25^3 to base 5

(b) 8^6 to base 2

(c) 9^4 to base 3

(d) 81^5 to base 3

Exercise

5. Determine the value of y :

$$16^{1/4} \times 2^y = 8^{3/4}$$

6. Evaluate:

(a) 10^0

(b) $81^{1/2}$

(c) $64^{1/3}$

(d) $25^{3/2}$

(e) $\frac{3^5 \times 3^2}{3^4}$

7. A patient has 4^3 affected cells on day 1. The number of affected cells doubles every day. The disease becomes serious when 2^{10} cells are affected. On which day does the disease become serious?
8. The area of a rectangle is $125^{1/4} \text{ cm}^2$. The side lengths are 5^{x+1} cm and $25^{1/2} \text{ cm}$. Determine x .

Solutions to Exercise 1.4.9

1. Evaluate:

(a) 7^3

$$7^3 = 7 \times 7 \times 7 = 343$$

(b) 3^7

$$3^7 = 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 = 2187$$

(c) 5^{-2}

$$5^{-2} = \frac{1}{5^2} = \frac{1}{25}$$

(d) $2^{5/2}$

$$2^{5/2} = (\sqrt{2})^5 = 4\sqrt{2}$$

or

$$2^{5/2} = \sqrt{2^5} = \sqrt{32} = 4\sqrt{2}$$

(e) $(-2)^6$

$$(-2)^6 = 64$$

because an even number of negative signs gives a positive result.

(f) -2^6

The index is applied first.

$$-2^6 = -(2^6) = -64$$

2. Express each as a single power:

(a) $2^3 \times 2^5 \times 2^7$

$$2^{3+5+7} = 2^{15}$$

(b) $3^3 \times 3^{-4}$

$$3^{3-4} = 3^{-1}$$

(c) $4^6 \times \frac{4^{-7}}{4^{-5}}$

$$\begin{aligned} 4^6 \times 4^{-7-(-5)} &= 4^6 \times 4^{-2} \\ &= 4^4 \end{aligned}$$

(d)

$$\frac{5^5 \times 5^{-3}}{5^4 \times 5^{-2} \times 5^{-7}}$$

Numerator:

$$5^{5-3} = 5^2$$

Denominator:

$$5^{4-2-7} = 5^{-5}$$

Therefore

$$\frac{5^2}{5^{-5}} = 5^{2-(-5)} = 5^7$$

(e) $(2^{-3})^4$

$$2^{-3 \times 4} = 2^{-12}$$

(f) $3^{2/3} \times 3^{-2}$

$$3^{2/3-2} = 3^{2/3-6/3} = 3^{-4/3}$$

3. Simplify:

(a) $y^5 \times y^8$

$$y^{5+8} = y^{13}$$

(b) $\frac{x^{10}}{x^7}$

$$x^{10-7} = x^3$$

(c) $x^3 \div x$

$$x^{3-1} = x^2$$

(d) $(a^2)^3$

$$a^{2 \times 3} = a^6$$

(e) $3x^2 \times 4x^7 \times 2x^{-3}$

Multiply coefficients:

$$3 \times 4 \times 2 = 24$$

Combine powers:

$$x^{2+7-3} = x^6$$

Therefore

$$24x^6$$

(f) $6x \times 7y^2 \times x^5$

$$42x^{1+5}y^2 = 42x^6y^2$$

4. Change to the specified base:

(a) 25^3 to base 5.

$$25 = 5^2$$

Therefore

$$25^3 = (5^2)^3 = 5^6$$

(b) 8^6 to base 2.

$$8 = 2^3$$

Therefore

$$8^6 = (2^3)^6 = 2^{18}$$

(c) 9^4 to base 3.

$$9 = 3^2$$

Therefore

$$9^4 = (3^2)^4 = 3^8$$

(d) 81^5 to base 3.

$$81 = 3^4$$

Therefore

$$81^5 = (3^4)^5 = 3^{20}$$

5. Determine the value of y :

$$16^{1/4} \times 2^y = 8^{3/4}$$

Write everything in base 2.

$$(2^4)^{1/4} \times 2^y = (2^3)^{3/4}$$

$$2 \times 2^y = 2^{9/4}$$

$$2^{1+y} = 2^{9/4}$$

Therefore

$$1 + y = \frac{9}{4}$$

$$y = \frac{9}{4} - 1 = \frac{5}{4}$$

6. Evaluate:

(a) 10^0

$$10^0 = 1$$

(b) $81^{1/2}$

$$\sqrt{81} = 9$$

(c) $64^{1/3}$

$$\sqrt[3]{64} = 4$$

(d) $25^{3/2}$

$$(\sqrt{25})^3 = 5^3 = 125$$

(e)

$$\begin{aligned}\frac{3^5 \times 3^2}{3^4} &= 3^{5+2-4} \\ &= 3^3 = 27\end{aligned}$$

7. Patient problem Initially:

$$4^3 = 64 = 2^6$$

The number doubles every day. Day 1:

$$2^6$$

Day 2:

$$2^7$$

Day 3:

$$2^8$$

Day 4:

$$2^9$$

Day 5:

$$2^{10}$$

The disease becomes serious on

Day 5

8. Rectangle problem Area

$$125^{1/4}$$

Side lengths:

$$5^{x+1} \quad \text{and} \quad 25^{1/2}$$

Simplify the second side:

$$25^{1/2} = 5$$

Therefore

$$5^{x+1} \times 5 = 125^{1/4}$$

Write everything in base 5.

$$\begin{aligned} 5^{x+1} \times 5^1 &= (5^3)^{1/4} \\ 5^{x+2} &= 5^{3/4} \end{aligned}$$

Hence

$$\begin{aligned} x + 2 &= \frac{3}{4} \\ x &= \frac{3}{4} - 2 \\ x &= \frac{3}{4} - \frac{8}{4} \\ x &= -\frac{5}{4} \\ x &= -\frac{5}{4} \end{aligned}$$